Simple average price relatives using A.M = P_{01} $= \dfrac{\sum \frac{p_1}{p_0} \times 100}{N}$

$= \dfrac{413.55}{4}$

$= 103.3875$

$P_{01} = 103.39$

The average price relative index number using arithmetic mean for the year 2012 is 103.39

Example 6.5

Construct simple average price relative index number using geometric mean for the year 2015 for the data showing the expenditure in education of the children taking different courses.

Expenditure per year	2014	2015
B.Sc	24000	26000
B.Com	20000	22000
B.E	108000	12000
M.B.B.S	150000	168000

Solution:

Expenditure	Year 2014 (p_0)	Year 2015 (p_1)	$P = (p_1/p_0) \times 100$	log P
B.Sc	24000	26000	$\frac{26000}{24000} \times 100 = 108.33$	2.0346
B.Com	20000	22000	$\frac{22000}{20000} \times 100 = 110.00$	2.0414
B.E	108000	12000	$\frac{120000}{108000} \times 100 = 111.11$	2.0457
M.B.B.S	150000	168000	$\frac{168000}{150000} \times 100 = 112.00$	2.0492
				$\sum \log P = 8.1709$

P_{01} = antilog $\left(\dfrac{\sum \log P}{N}\right)$

= antilog $\left(\dfrac{8.1709}{4}\right)$

= antilog (2.04275)

= antilog (2.0428)

= 110.4

The average price relative index number using geometric mean for the year 2015 is 110.4

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